

RESEARCH ARTICLE | DECEMBER 09 2022

Classification of hyperchaotic, chaotic, and regular signals using single nonlinear node delay-based reservoir computers

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Chaos 32, 123126 (2022)

<https://doi.org/10.1063/5.0124204>



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Cite as: Chaos 32, 123126 (2022); doi: 10.1063/5.0124204

Submitted: 4 September 2022 · Accepted: 21 November 2022 ·

Published Online: 9 December 2022



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ABSTRACT

The Lyapunov exponent method is generally used for classifying hyperchaotic, chaotic, and regular dynamics based on the equations modeling the system. However, several systems do not benefit from appropriate modeling underlying their dynamic behaviors. Therefore, having methods for classifying hyperchaotic, chaotic, and regular dynamics using only the observational data generated either by the theoretical or the experimental systems is crucial. In this paper, we use single nonlinear node delay-based reservoir computers to separate hyperchaotic, chaotic, and regular dynamics. We show that their classification capabilities are robust with an accuracy of up to 99.61% and 99.03% using the Mackey–Glass and the optoelectronic oscillator delay-based reservoir computers, respectively. Moreover, we demonstrate that the reservoir computers trained with the two-dimensional Hénon-logistic map can classify the dynamical state of another system (for instance, the two-dimensional sine-logistic modulation map). Our solution extends the state-of-the-art machine learning and deep learning approaches for chaos detection by introducing the detection of hyperchaotic signals.

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In dynamical systems, the quantifiers for classifying hyperchaotic, chaotic, and regular signals are potent tools since they allow one to characterize the dynamics of a given system. Calculating most of these quantifiers requires the equations modeling the system under investigation. However, the models governing the system are not always affordable in many situations where complex dynamics may occur, such as in astronomy, oceanography, meteorology, and satellite operations. In these situations, predicting a trajectory or motion using only the system data is crucial. This work uses delay-based reservoir computers to classify hyperchaotic, chaotic, and regular signals based only on the system data. We achieve quasi-maximal accuracies using the Mackey–Glass and the optoelectronic oscillator (OEO) delay-based reservoir computers. This classification can be used to make operational decisions in applications where distinguishing hyperchaotic, chaotic, and regular dynamics is required.

I. INTRODUCTION

Many natural systems, as well as manufactured devices, exhibit chaotic and non-chaotic signals. These systems include, for instance, heartbeat irregularities, weather, climate, the stock market, and road traffic, among others. For some systems, chaotic behavior is an undesired phenomenon devoted to the control theory. However, for others, chaotic signals are important for research and technological purposes. For instance, chaos theory is used in a wide range of research and engineering fields, such as secure communication,^{1–3} chaotic radar,^{4,5} random numbers generation,^{6–8} digital signature application,⁹ and more.¹⁰

Therefore, the need to classify chaos from regular dynamics, such as periodic and quasi-periodic dynamics, is usually investigated using quantifiers, such as phase portrait, power spectrum, Poincaré section, Lyapunov exponents (LE_s), and entropy.¹¹ Recently, novel quantifiers based on deep learning methods have been explored

to classify chaotic from regular dynamics.^{12–14} For instance, the classification accuracies of 94.4% and 98% were independently obtained using convolutional neural networks when training the model with one variable of the Lorenz system and the standard map, respectively.^{12,13} Furthermore, Celletti *et al.* extended the findings presented in Ref. 13 to classify non-chaotic and chaotic motions for the specific problem of celestial mechanics using deep learning methods.¹⁴ The authors studied different deep learning architectures and showed that deep learning models based on convolutional neural networks (Resnet, XCM, InceptionTime) outperform other architectures, such as the long short-term memory model. In addition, they showed that the InceptionTime network outperforms all the other architectures with an accuracy of 99%.¹⁴

Since these deep learning methods are challenging to train, time-consuming, and computationally expensive, in our previous work, we introduced the use of the machine learning method based on reservoir computing to classify chaotic and non-chaotic signals.¹⁵ Using a Mackey–Glass oscillator as the single nonlinear node (NLN) delay-based reservoir computer, we showed that when trained with the standard map, the Mackey–Glass reservoir computer is robust and reliable in classifying chaotic and non-chaotic signals with a classification accuracy of up to 99.03% with short time series. In addition, we demonstrated that the reservoir computer trained with the standard map could classify chaotic and non-chaotic signals of the Lorenz system.¹⁵

However, none of these deep learning methods presented in Refs. 12–14 and the machine learning method¹⁵ dealt with the classification of hyperchaotic dynamics from other dynamics. Yet, it is necessary since hyperchaos presents several advantages over chaos. Hyperchaotic dynamics are fundamentally more complex than chaotic dynamics. Therefore, hyperchaos is more appropriate and effective for many applications, such as cryptography, secure communication, entropy source for random number generators, and more.^{8,16–18}

In this paper, we implement a machine learning method known as single nonlinear node delay-based reservoir computing to distinguish hyperchaotic from chaotic and regular behaviors. This method is advantageous over the deep learning methods because it requires only a single nonlinear node, is time and energy-efficient,^{19–28} and can allow a neuromorphic implementation.^{20,21,24,29–32}

The Mackey–Glass oscillator (MGO) and the optoelectronic oscillator (OEO) are the two distinct single nonlinear node delay-based reservoir computers used. Our interest is focused on these two reservoir computers because: First, the MGO delay-based reservoir computer is easy to implement in hardware and has been successfully used for several machine learning tasks such as spoken digit recognition, sunspot prediction, time series identification, electrocardiogram classification, electrocardiogram prediction, chaotic attractor reconstruction, and more.^{23,33–35} Moreover, the MGO reservoir computer is cheaper and implementable with commercial electronic components, making it easy to cascade several to form deep delay-based reservoir computers.^{26,35,36} Second, the OEO delay-based reservoir computer can operate faster (up to 100 GHz) with the natural ability to perform high-speed signal processing.^{28,37} For instance, it has been successfully used in several time series classification, and prediction tasks, such as signal recognition, IQ-modulated signals classification, spoken digit

recognition, sunspot prediction, and more.^{27,28,38–41} For each reservoir computer, two cases are analyzed: in the first case, the training and testing datasets are from the two-dimensional Hénon-logistic map (2D-HLM). In the second case, the training dataset is generated by the 2D-HLM, while the testing dataset comes from the two-dimensional sine-logistic modulation map (2D-SLMM).

The paper is organized as follows. Section II describes the labeling method of regular, chaotic, and hyperchaotic trajectories. Section III presents the architecture of the reservoir computing and its nonlinear nodes. Section IV contains the results. Specific machine learning evaluation measures are presented. Section V is devoted to a conclusion.

II. BUILDING THE DATASET

The system used to build the dataset is the two-dimensional Hénon-logistic map (2D-HLM). This map derives from the 2D Hénon map and the one-dimensional (1D) logistic map, and it can exhibit regular, chaotic, and hyperchaotic behaviors.⁴² The following 2D equation describes its dynamics:⁴²

$$\begin{cases} x_{n+1} = a + by_n(1 - cy_n) - ax_n^2, \\ y_{n+1} = c^2 y_n(1 - by_n), \end{cases} \quad (1)$$

with n being an integer. Here, the parameters are $a = 0.895$ and $b = 8$. The control parameter is $c \in [1, 2]$. While varying c , it is demonstrated that this 2D-HLM displays three dynamics from regular $c \in [1.0, 1.74]$, chaotic $c \in [1.74, 1.88]$, to hyperchaotic $c \in [1.88, 2.0]$ (see Fig. 1). The Lyapunov exponents (LE_s) λ_1 and λ_2 of the 2D Hénon-logistic map for each dimension are computed using the following equations:^{43,44}

$$\lambda_{1,2} = \lim_{n \rightarrow \infty} \frac{1}{n} \ln(v_{1,2}(n)), \quad (2)$$

with $v_{1,2}(n)$ being the eigenvalues of the matrix Q_n built from the product of the Jacobian matrices J_k of Eq. (1) and defined as

$$Q_n = \prod_{k=0}^{n-1} J_k. \quad (3)$$

The precision of the LE_s method increases with the length of the time series. Therefore, to obtain high precision, we use a long simulation time of 3×10^5 to compute λ_1 and λ_2 as in Ref. 45. Based on λ_1 and λ_2 , 2601 values of the control parameter c are randomly selected and labeled into three classes: those generating regular (labeled regular), chaotic (labeled chaos), and hyperchaotic (labeled hyperchaos) behaviors. The Lyapunov exponent diagram of Fig. 1 illustrates this separation. Indeed, regular dynamics are characterized by all negative or zero Lyapunov exponents. Chaos is described by one positive Lyapunov exponent, while at least two positive Lyapunov exponents identify hyperchaos⁴³ (see Fig. 1). Each of the 2601 randomly selected control parameters values c is used to generate a trajectory of length 100 after having discarded a transient of 600. Each trajectory is labeled with the dynamic regime corresponding to the control parameter. These trajectories form the dataset and are split into 2081 training and 520 testing trajectories.¹³ The training set contains 694 regular, 694 chaotic, and 693 hyperchaotic trajectories, while the testing set contains 173 regular, 173 chaotic, and 174

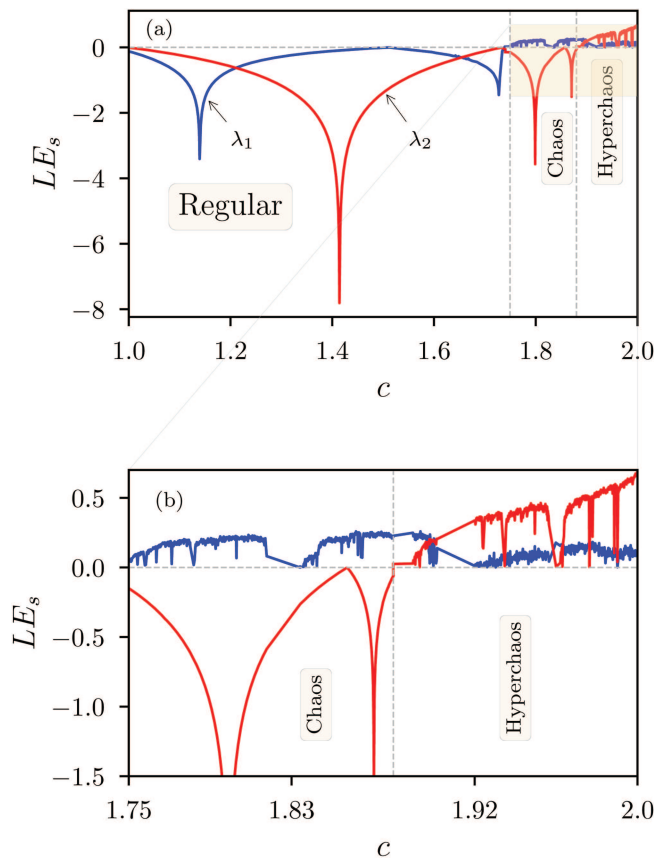


FIG. 1. Plot of the Lyapunov exponents vs the control parameter c for the two-dimensional Hénon-logistic map (2D-HLM). The Lyapunov exponents are λ_1 (blue) and λ_2 (red). The boxes regular, chaos, and hyperchaos indicate the ranges of the control parameter where the system depicts regular, chaotic, and hyperchaotic dynamics, respectively. The simulation time is 3×10^5 and the initial conditions are taken as $x_0 = y_0 = 0.1$. (b) is the magnification of (a) in the chaotic and hyperchaotic regions.

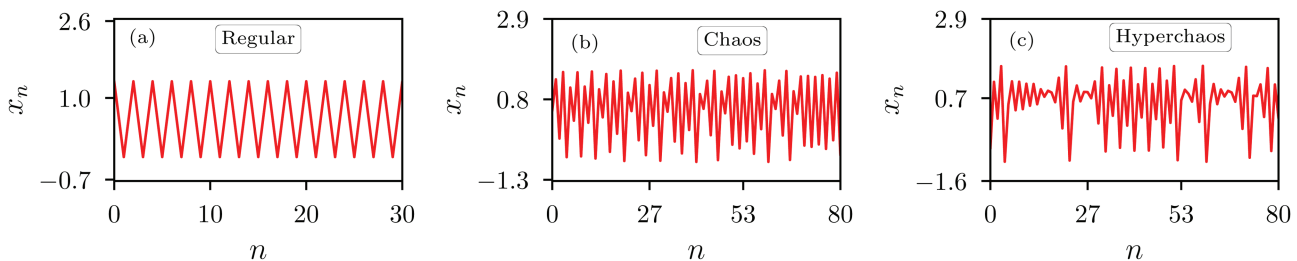


FIG. 2. Time series of some labeled trajectories of Fig. 1. The chaotic and hyperchaotic time series develop as disorder trajectories, while the regular ones do not. The control parameters c for the time series (a)–(c) are 1.5, 1.8, and 1.9, respectively. The Lyapunov exponents (λ_1, λ_2) of (a)–(c) are $(-0.0075, -1.3847)$, $(0.2061, -2.0178)$, and $(0.1433, 0.2076)$, respectively. The initial conditions for the time series (a)–(c) are the same and equal to the couple $(1.0, 1.0)$.

hyperchaotic trajectories. Some examples of these trajectories are displayed in Fig. 2. One can notice that chaotic and hyperchaotic time series develop as disorder trajectories compared to the regular ones. However, this visual distinction does not help to differentiate chaotic from hyperchaotic behaviors. In Sec. III, we introduce the novel machine learning classification method using the neuromorphic architecture known as the single nonlinear node delay-based reservoir computing.

III. SINGLE NONLINEAR NODE DELAYED-BASED RESERVOIR COMPUTING ARCHITECTURE

Once the dataset is constructed, the neuromorphic architecture of Fig. 3 is used to implement the single nonlinear node delay-based reservoir computing method. As illustrated in Fig. 3, our reservoir computer has three layers: the input layer for masking the input sample $u(t)$, the reservoir layer for processing the input with the nonlinear node (NLN), and the output layer (readout) for adapting the output weights through the learning process. Throughout the paper, a two-dimensional random matrix of binary values (-1 and 1) serves as the input mask.^{23,46} The reservoir is configured as a ring with only one active nonlinear node (NLN). The additional nodes (virtual nodes) represent the delayed output of that nonlinear node (NLN).

In this study, we consider two single nonlinear node delay-based reservoir computers. The first is the Mackey–Glass oscillator (MGO), and the second is the optoelectronic oscillator (OEO). The MGO and OEO delay-based reservoir computers are given respectively as follows:³³

$$\dot{x}(t) = -x(t) + \frac{\eta[x(t-\tau) + \gamma I(t, n)]}{1 + [x(t-\tau) + \gamma I(t, n)]^p}, \quad (4)$$

$$\dot{x} = -x(t) + \frac{1}{\Omega} \int_{-\infty}^t x(s) ds + \eta \sin^2(x(t-\tau) + \gamma I(t, n) + \phi), \quad (5)$$

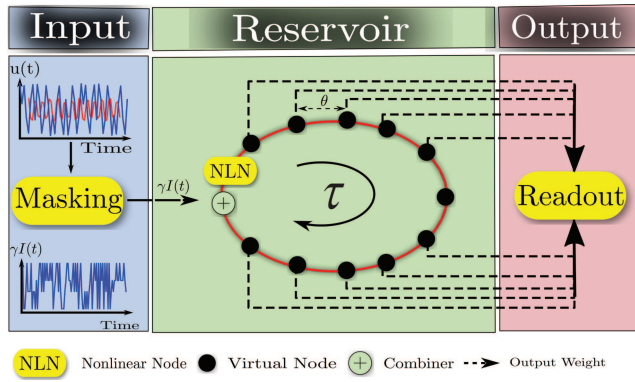


FIG. 3. Architecture of the single nonlinear node (NLN) delayed-based reservoir computer. In this work, the nonlinear node (NLN) is either the Mackey–Glass oscillator (MGO) or the optoelectronic oscillator (OEO).

where t is the time. $x(t)$ and $x(t - \tau)$ represent the dynamical variables and delayed variables, respectively. $I(t, n)$ is the masked input signal. The parameters η and γ denote the feedback strength and the input scaling, respectively. Noting that the nonlinear exponent p of Eq. (4) can be modified to adapt to a given machine learning task.^{21,33,47,48} In Eq. (5), Ω is the high and low-pass filter timescales ratio and ϕ is the phase. For computational convenience and setting $y = \frac{1}{\Omega} \int_{-\infty}^t x(s) ds$, Eq. (5) can be written in the form of a flow of first-order coupled equations,

$$\begin{cases} \dot{x} = -x(t) + \frac{\gamma}{\Omega} + \eta \sin^2(x(t - \tau) + \gamma I(t, n) + \phi), \\ \dot{y} = x. \end{cases} \quad (6)$$

For both reservoir computers, each trajectory is identified with numeric indexes (for instance, 0, 1, and 2 for regular, chaotic, and hyperchaotic, respectively). Y is the output or target matrix containing the target indexes (0, 1, or 2). The input dataset then consists of 2601 time series (x_n, y_n), and the system's output is the target matrix mapping the actual (true) dynamics for each of the 2601 time series inputs. Each time series (x_n, y_n) from the dataset is injected into the reservoir to create an intermediate state matrix containing reservoir responses for that time series. For this purpose, each input sample $u(t)$ of the regular, chaotic, or hyperchaotic signal to be processed is held during one time-delay τ . This sample is then multiplied with the input mask to create the masked input signal $\gamma I(t)$ (see Fig. 3). The signal $\gamma I(t)$ is injected into the nonlinear node after each time interval of θ to keep the reservoir in a transient state rather than a steady state. This injection process at θ interval emulates a neural network of $M = \tau/\theta$ virtual nodes. After injecting all the sample points of a trajectory (regular, chaotic, or hyperchaotic), the reservoir response is collected to form a state matrix. The state matrices for all trajectories of the dataset are then brought together into a unique matrix, X . The X matrix and its corresponding target of indexes matrix Y are used for the readout process based on the ridge regression (or Thikonov regularization), which permits to compute

TABLE I. List of the parameters of the MGO and OEO delay-based reservoir computers with their corresponding values. With the exception of the input scaling, these parameters are compatible with the experimental values.^{21,33} For the MGO and OEO, the ridge regression coefficients α are 1.0×10^{-10} and 0.9, respectively.

Parameters	Labels	Values
Input scaling	γ	0.5
Injection interval	θ	0.2
Feedback strength	η	0.55
Number of virtual nodes	M	400
Nonlinear exponent	p	7
Phase	ϕ	$3\pi/4$
Filter timescales ratio	Ω	100

the output weights matrix (W_{out}) as follows:^{49,50}

$$W_{out} = (YX^T)(XX^T + \alpha \mathbb{1})^{-1}, \quad (7)$$

where $\mathbb{1}$ is the identity matrix, T stands for the transpose, and α is the regularization parameter also known as the Ridge regression coefficient and is used to minimize the magnitude of the weights of W_{out} , mitigating the sensibility to noise and over-fitting. A classification of the reservoir output is then obtained by using the output weights W_{out} to assign a dynamical behavior (regular, chaos, or hyperchaos) to every new input signal $u(t)$ based on the winner-takes-all principle commonly used in reservoir computing systems.^{38,51} The results of the training and testing sets are exposed in Sec. IV.

IV. RESULTS AND DISCUSSIONS

In this section, except the input scaling and ridge regression coefficient, the complete list of the parameters is taken to be compatible with the experimental values and is given in Table I.

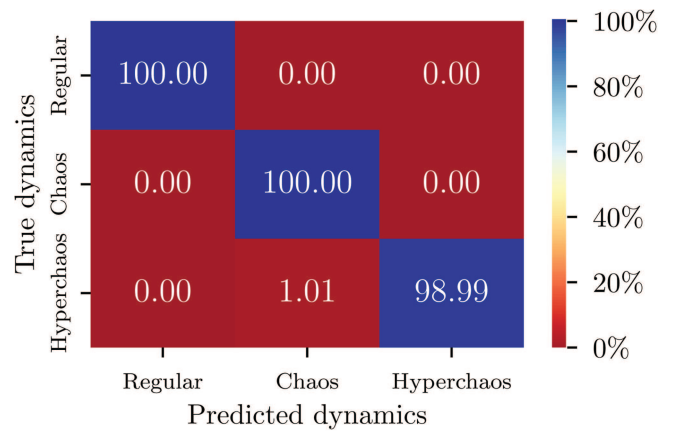


FIG. 4. Confusion matrix of all the 2081 training trajectories when using the MGO delay-based reservoir computer. Parameters are those of Table I.

TABLE II. Classification report of the 2081 training trajectories when using the MGO delay-based reservoir computer. Parameters are those of Table I.

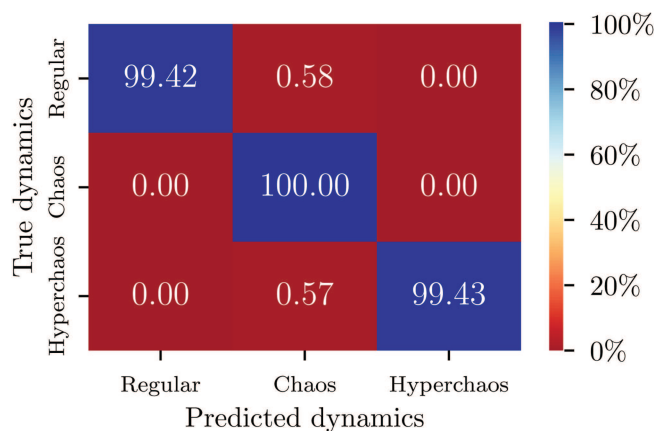
Dynamics	Precision (%)	Recall (%)	f_1 -score (%)
Regular	100	100	100
Chaos	99.00	100	99.49
Hyperchaos	100	98.98	99.49
f_1 macro avg-score (%)			99.66
Accuracy (%)			99.66

A. Recognition performance using the MGO delay-based reservoir computer

1. Results of the training set with the MGO delay-based reservoir computer

The confusion matrix of the training set when using the MGO delay-based reservoir computer is presented in Fig. 4. Each row represents the actual dynamics (true dynamics), while each column represents the predicted dynamics. From Fig. 4, one can see that the percentage of trajectories that are correctly classified (on the diagonal) is largely superior to the percentage of misclassified trajectories (off diagonal). However, only 1.01% of hyperchaotic trajectories are misclassified as chaotic trajectories. Interestingly, all the regular and chaotic trajectories are correctly classified (see Fig. 4).

Moreover, the classification report is done and Table II gives the result. This table shows that the f_1 -scores are 100%, 99.49%, and 99.49% for regular, chaotic, and hyperchaotic dynamics, respectively. The system also performs the training set with a f_1 macro avg-score of 99.66% and an accuracy of 99.66%. These results indicate that the MGO delay-based reservoir computer is well trained and can be used for testing.

**FIG. 5.** Confusion matrix of the 520 testing trajectories when using the trained MGO delay-based reservoir computer. Parameters are those of Table I.**TABLE III.** Classification report of the 520 testing trajectories when using the trained MGO delay-based reservoir computer. Parameters are those of Table I.

Dynamics	Precision (%)	Recall (%)	f_1 -score (%)
Regular	100	99.42	99.71
Chaos	98.85	100	98.42
Hyperchaos	100	99.42	99.71
f_1 macro avg-score (%)			99.61
Accuracy (%)			99.61

2. Results of the testing set with the MGO delay-based reservoir computer

The resulting weights from the MGO delay-based reservoir computer training are now used for the testing. The result is displayed on the confusion matrix of Fig. 5. This figure shows that only 0.58% of regular and 0.57% of hyperchaotic testing trajectories are misclassified as chaotic trajectories. However, all the chaotic trajectories are perfectly classified (see Fig. 5). In addition, the corresponding classification report shown in Table III presents a f_1 -score of 99.71%, 98.42%, and 99.71% for regular, chaotic, and hyperchaotic dynamics, respectively. Furthermore, the system performs with a f_1 macro avg-score, and an accuracy achieving 99.61% (see Table III).

B. Recognition performance with the OEO delay-based reservoir computer

1. Results of the training set with the OEO delay-based reservoir computer

With the OEO delay-based reservoir computer, the confusion matrix of the training set is depicted in Fig. 6. It appears that the percentage of trajectories that are correctly classified (on the diagonal) is also largely superior to the percentage of misclassified trajectories (off diagonal). Only 0.14% of chaotic trajectories and 1.30% of hyperchaotic trajectories are misclassified as regular trajectories.

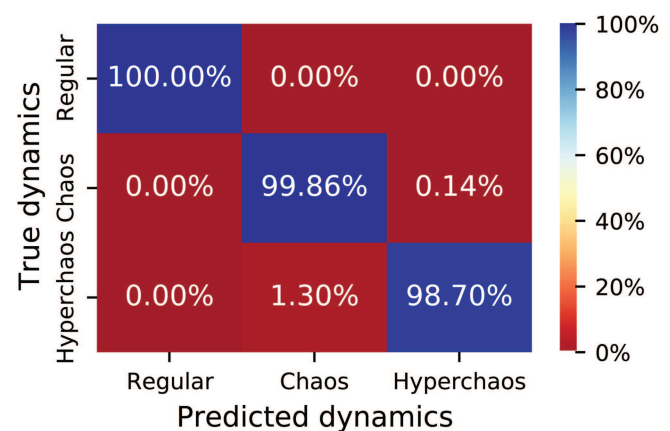
**FIG. 6.** Confusion matrix of the 2081 training trajectories when using the OEO delay-based reservoir computer. Parameters are those of Table I.

TABLE IV. Classification report of the 2081 training trajectories when using the OEO delay-based reservoir computer. Parameters are those of Table I.

Dynamics	Precision (%)	Recall (%)	f_1 -score (%)
Regular	100	100	100
Chaos	98.71	99.85	99.28
Hyperchaos	99.85	98.70	99.27
f_1 macro avg-score (%)			99.51
Accuracy (%)			99.51

of hyperchaotic trajectories are misclassified as hyperchaotic and chaotic trajectories, respectively. However, all the regular trajectories are perfectly classified (see Fig. 6). Moreover, the classification report shows a f_1 -score of 100%, 99.28%, and 99.27% for regular, chaotic, and hyperchaotic dynamics, respectively (see Table IV). This table also shows that the system performs the training set with a f_1 macro avg-score of 99.51% and an accuracy of 99.51%. These results again demonstrate the successful training of the OEO delay-based reservoir computer, which is then ready for testing.

2. Results of the testing set with the OEO delay-based reservoir computer

Figure 7 shows the confusion matrix of testing when using the OEO as a reservoir computer. This figure shows that the percentage of correctly classified trajectories (on the diagonal) is still huge compared to the percentage of misclassified trajectories (off diagonal). For each dynamics, more than 98% of trajectories are correctly classified (see Fig. 7). In addition, the classification report of the testing set given in Table V demonstrates that the f_1 -scores are up to 99.71%, 98.56%, and 98.84% for regular, chaotic, and hyperchaotic dynamics, respectively. The system also performs with a f_1 macro avg-score of 99.04% and an accuracy achieving 99.03%.

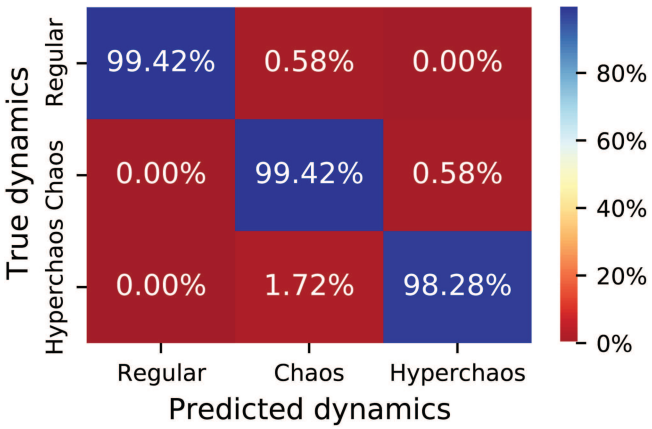


FIG. 7. Confusion matrix of the 520 testing trajectories when using the trained OEO delay-based reservoir computer. Parameters are those of Table I.

TABLE V. Classification report of the 520 testing trajectories when using the OEO delay-based reservoir computer. Parameters are those of Table I.

Dynamics	Precision (%)	Recall (%)	f_1 -score (%)
Regular	100	99.42	99.71
Chaos	97.72	99.42	98.56
Hyperchaos	99.41	98.27	98.84
f_1 macro avg-score (%)			99.04
Accuracy (%)			99.03

C. Performance of the MGO and OEO delay-based reservoir computers trained with the Hénon-logistic map (HLM) and tested with the sine-logistic modulation map (SLMM)

Based on the idea developed in Refs. 12–15, the purpose of this

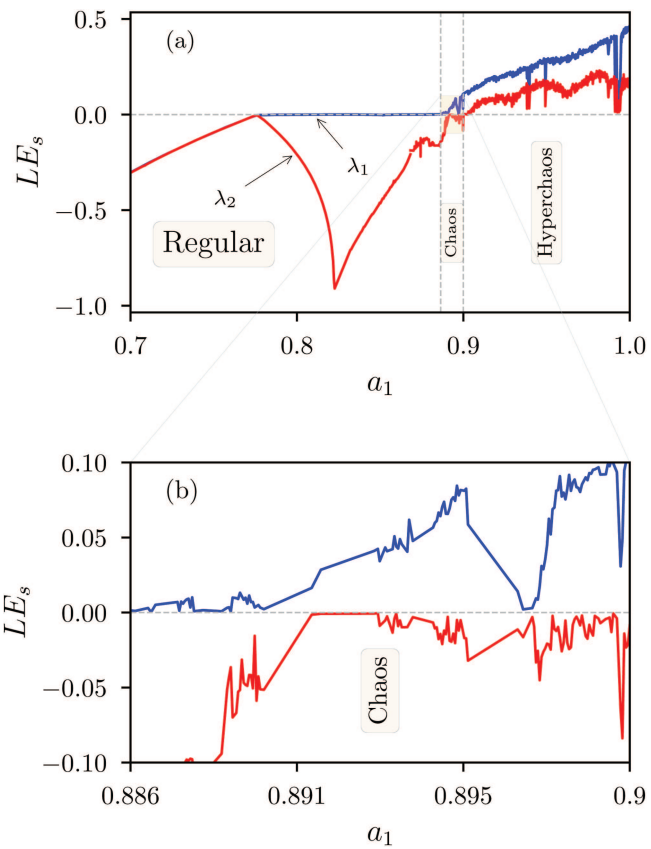


FIG. 8. (a) Lyapunov exponents diagram of the sine-logistic modulation map (2D-SLMM). The LE_s with blue, red, and black colors corresponds to the control parameter a_1 values labeled regular, chaotic, and hyperchaotic dynamics. Here, $\mu = 1.0$, $b_1 = 3$ and the initial conditions equal the couple (0.1, 0.1). The simulation time is 3×10^5 . The generated trajectories are used to evaluate the MGO, and the OEO delay-based reservoir computers, trained with time series from the 2D Hénon-logistic map. (b) is the magnification of (a) in the chaotic domain.

TABLE VI. Predictions performance for the 2D sine-logistic modulation map (2D-SLMM) with the reservoir computers trained with the 2D Hénon-logistic map. Parameters are those of Table I.

System	Dynamics	Precision (%)	Recall (%)	f_1 -score (%)
MGO	Regular	95.00	19.00	31.00
	Chaos	46.73	86.00	60.00
	Hyperchaos	85.41	82.00	83.67
	f_1 macro avg-score (%)			62.33
	Accuracy (%)			62.33
OEO	Regular	76.44	98.13	85.94
	Chaos	60.56	45.74	52.12
	Hyperchaos	85.54	70.14	77.08
	f_1 macro avg-score (%)			71.71
	Accuracy (%)			77.30

subsection is to evaluate the performance of the two delay-based reservoir computers to classify the dynamical behaviors of another system such as the sine-logistic modulation map (2D-SLMM) when they are trained with the 2D Hénon-logistic map (2D-HLM). The 2D-SLMM has a different nonlinearity than the 2D-HLM and consists of the following set of equations:^{42,52}

$$\begin{cases} x_{n+1} = a_1(\sin((\pi\mu y_n) + b_1)x_n(1 - x_n)), \\ y_{n+1} = a_1(\sin((\pi\mu x_{n+1}) + b_1)y_n(1 - y_n)). \end{cases} \quad (8)$$

The parameter a_1 is defined in the range values $a_1 \in [0, 1]$. Other parameters are $\mu = 1.0$ and $b_1 = 3$. With these parameters, the 2D-SLMM displays the three dynamical behaviors as shown in Fig. 8. We apply the sampling technique described in Sec. II to separate, generate, and label the trajectories. The generated testing data consist of 520 trajectories. The MGO and OEO delay-based reservoir computers perform the 2D-SLMM testing data with a classification accuracy of 62.33% and 77.30%, respectively (see Table VI). One can notice that the two delay-based reservoir computers can still predict the dynamical states of the 2D-SLMM with relatively good classification accuracies though lower than the values found when they are trained and tested with data from the same 2D-HLM (99.61% and 99.03% of Tables III and V). A similar decrease was also obtained when using deep learning methods to classify regular and chaotic dynamics.^{12–14}

V. CONCLUSION

In this paper, we have proposed a single nonlinear node delay-based reservoir computing approach to classifying three dynamical states, namely, regular, chaos, and hyperchaos. We have used the MGO and OEO delay-based reservoir computers and have investigated two training-testing scenarios. In the first scenario, the reservoir computers have been trained and tested with data from the same system, the 2D-HLM. The results show that the MGO and OEO delay-based reservoir computers can recognize whether a dynamics is regular, chaotic, or hyperchaotic with an accuracy greater than 99%. These high scores demonstrate that these reservoir

computers are reliable and do not contain structural weaknesses in the reported scores. In the second scenario, the reservoir computers have been trained with data from the 2D-HLM and tested with data from another system having a different type of nonlinearity, the 2D-SLMM. Interestingly, it results in a classification accuracy of up to 62.33% and 77.30% using the MGO and OEO delay-based reservoir computers, respectively. Furthermore, from our results, in contrast to the state-of-the-art machine learning and deep learning approaches^{12–15} that deal with regular and chaotic behaviors, our MGO and OEO delay-based reservoir computers can additionally detect hyperchaos.

Our delay-based reservoir computing method for detecting regular, chaotic, and hyperchaotic dynamics does not necessarily require knowledge of the dynamical equation describing the system from which the dynamics are to be classified. Instead, it is based entirely on any available data set. This method has the additional advantage of providing real-time experimental facilities and meets the future requirements of embedded terrestrial, marine, and space systems analysis. A potential extension of this work could be investigating the understanding and improvement of the results in the second scenario. This extension could contribute to a step toward universal dynamical state quantifiers based on reservoir computers. Another further investigation could be carrying the classification developed in the framework of this paper in the presence of noise.⁵³ One can also extend by including the method of surrogate data in the machine learning process, for instance, during testing.⁵⁴

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author contributions

All the authors were involved in the preparation of the manuscript. All the authors have read and approved the final manuscript.

Dagobert Wenkack Liedji: Conceptualization (lead); Formal analysis (lead); Investigation (lead); Writing – original draft (equal); Writing – review & editing (supporting). **Jimmi Hervé Talla Mbé:** Formal analysis (equal); Project administration (equal); Writing – original draft (equal); Writing – review & editing (lead). **Godpromesse Kenne:** Project administration (lead); Supervision (lead); Writing – review & editing (supporting).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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